Nexus Architecture Atlas

This walkthrough maps Nexus's durable user flows to their owning components.

User flow	Primary owner	Supporting contract	Evidence
Install or repair media	`scripts/installer/`	Owner-aware lease; atomic runtime state	Tests; packaged run
Generate an image	Image dispatcher and Python pipelines	Shared readiness API	Validated PNG artifact
Edit or recover an image	Image edit runtime and viewer	Revisioned recipes; approval recovery; retained originals	Retry, restart, restore and grant tests
Generate a video	Video dispatcher and Python pipelines	Shared readiness API	Probed video artifact
Browse and install models	`core/registry/` and Settings	Shared availability/recommendation tuple	Cross-language rank fixture
Read chat usage	`core/chat/` and shared chat UI	Role-local token provenance	Protocol, hydration, and UI tests
Organize chats; select workspaces	Chat explorer and Coding workspace	Persisted folder colors; union-of-roots boundary	Database reopen, picker and denial tests

End-to-end path

1. Catalog-source reachability checks and desktop payload preparation are packaging prerequisites. The installer provisions the media runtime.
2. The sidecar reads the atomic readiness record and exposes one repair contract to Image Studio and Video Lab.
3. Generation jobs check the selected model directory layout before loading models and finalize artifacts before reporting success.
4. The desktop renders truthful capability, model, transcript, archive, and workspace state from shared contracts.
5. Packaged acceptance requires installing the exact candidate, retaining the installer-to-payload binding, and exercising the installed runtime with the media smoke harness. The harness alone does not establish

that binding.

Release-specific logs and test counts live in the corresponding release or archive tree.